

Yang Zhang

AI Agent Engineering · LLM Applications · Data Science · Quantitative Finance
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andyzyoung.github.io

EDUCATION

University of Michigan, Ann Arbor

Master of Science in Quantitative Finance and Risk Management

Sep 2026 – Jun 2028 (Expected)

Shanghai University

Bachelor of Engineering in Data Science and Big Data Technology

Sep 2022 – Jun 2026 (Conferred)

GPA: 3.57/4.00

PROFESSIONAL EXPERIENCE

EY Ernst & Young AI Agent / Full-Stack Developer Intern

Mar 2026 – Jun 2026

Built enterprise LLM applications for AI-news intelligence, agentic retrieval, Outlook email analysis, and Teams meeting automation.

- Built an AI-news intelligence **AI agent** with a self-built **ReAct agent runtime** and **parallel multi-agent** orchestration, where topic-specific CollectorAgents used **OpenAI API function calling** to select RSS and web-search tools and collect articles concurrently, with downstream scoring, URL verification, and real-time execution tracing delivered through **FastAPI** and **Server-Sent Events**.
- Implemented the agent's retrieval and memory modules with **OpenAI Embeddings API**, **BM25**, vector search, hybrid retrieval, semantic deduplication, and **ChromaDB**; persisted articles, **RAG traces**, agent events, and deduplication decisions in **SQLite** through **SQLAlchemy ORM**, and exposed reusable tools through **MCP** with service packaging via **Docker**.
- Developed a **Python backend service** for an **Outlook Add-in** hosted on **Microsoft Azure**; invoked the **OpenAI API** and **Microsoft Graph API** to summarize emails, generate reply drafts, extract meeting details, identify participants, query common availability, and create **Teams** meetings.
- Configured **Microsoft Entra ID**, **OAuth 2.0/OIDC**, token refresh, and **Microsoft Graph delegated permissions**; matched attendees against organization users, queried multi-party availability, and created **Teams** meetings from POC to dev deployment and InfoSec testing.

China UnionPay Data Co., Ltd. Data Analyst Intern

Jul 2024 – Aug 2024

Automated credit-card operating reports and bank-level benchmarking for monthly business analysis.

- Processed SQL-derived BI exports and Excel workbooks with **Python**, **Pandas**, **NumPy**, **Matplotlib**, and **Power BI**; built dashboards for active cards, overdraft balances, utilization, customer segments, installment products, and campaign performance.
- Built reusable monthly and annual benchmarking logic across **9 partner banks** and **16 listed banks**, covering month-over-month trends, account-age curves, customer-value segments, credit limits, non-performing balances, delinquency ratios, transaction volumes, and complaint ratios.

PROJECT & RESEARCH EXPERIENCE

Weakly Supervised Video Anomaly Detection Computer Vision / Deep Learning Engineer

Senior Thesis, 2026

Built a surveillance-video anomaly detection system that scores abnormal events from video-level labels and supports web-based inference.

- Built the weakly supervised anomaly detection workflow on the **UCF-Crime** dataset; converted surveillance videos into **32 x 2048** segment features with a **3D-ResNet-101** feature extractor pretrained on Kinetics/KM.
- Implemented **PyTorch** detector variants with **MIL ranking loss**, **Dense BCE**, magnitude-based **top-k** segment selection, **multi-scale dilated temporal Conv1d**, LayerNorm, dropout, feature-magnitude contrast, and memory/temporal-attention experiments.
- Built training and evaluation tooling with **TensorBoard**, **AdamW**, cosine learning-rate scheduling, gradient clipping, **EMA**, seed search, checkpoint scanning, and frame-level **ROC/AUC**; improved single-model AUC to approximately **0.8366** and greedy-ensemble AUC to approximately **0.8479**.
- Developed an inference and web-demo workflow with **Python**, **PyTorch**, **OpenCV**, and **NumPy**, supporting uploaded videos or feature files and mapping 32 segment scores back to frame/time ranges.

AI-Powered Course Assistant System Full-Stack LLM Developer

Feb 2025 – May 2025

Built an AI tutor that combines retrieval-augmented Q&A, knowledge-graph retrieval, course management, and a web-based digital human.

- Built the local course-material Q&A backend with **Ollama**, **DeepSeek-R1**, and **BGE-Large-ZH-v1.5 embeddings**; embedded lecture materials and returned grounded answers for teaching scenarios instead of general-purpose LLM responses.
- Implemented graph-based retrieval with **LightRAG**, including entity/relation extraction, low-level entity retrieval, high-level relationship retrieval, incremental knowledge-base updates, and graph visualization with **NetworkX**, **Pyvis**, and **Streamlit**.
- Developed a course-management web interface with **Vue 3**, **Axios**, and **HTML/CSS/JavaScript**; connected course cards, video pages, Q&A results, and knowledge-graph views to backend APIs.
- Integrated a digital-human interface with **Three.js**, **Blender** assets, **AnimationMixer**, and **Microsoft Azure TTS**; fixed model loading, data-format, integration, and animation timing issues while maintaining **30 FPS** rendering and sub-**3s** responses in project tests.

MindSpore Innovation Camp NLP / LLM Engineer

Mar 2024 – Apr 2024

Completed hands-on NLP and LLM deployment exercises using MindSpore, Transformer models, BERT, ChatGLM, and Gradio.

- Trained and evaluated **Transformer** and **BERT** models with **MindSpore**, **MindNLP**, Huawei Cloud **ModelArts**, and an **NVIDIA P100 GPU**, covering sequence modeling, sentiment classification, inference, and BLEU-score evaluation.
- Deployed a local **ChatGLM-6B** chatbot through **Gradio**; studied **PEFT**, prompt tuning, instruction tuning, and LLM application workflows, and received Second Prize in the camp.

PUBLICATION

Customer Segmentation: Data-Driven Research on Transaction-Level Data of JD.com First Author

ICCSMT 2024, EI-indexed

Built a customer-preference modeling study on JD.com transaction, click, user-profile, SKU, delivery, and inventory data.

Processed **JD.com transaction-level dataset** with **Python**, **Pandas**, and **NumPy**; engineered SKU/user-preference features from discount, final price, product attributes, city level, PLUS membership, and click behavior, then compared **K-means** segmentation with **Logistic Regression**, **Decision Tree**, **Random Forest**, and **Stacking** models using **RFE**, **Grid Search**, and **Randomized Search**.

SKILLS

Programming: Python, C/C++, SQL, JavaScript, HTML/CSS, MATLAB

AI / LLM / Agents: AI agent, GPT/OpenAI API, ReAct agent runtime, tool calling, RAG, embeddings, BM25/vector/hybrid retrieval, semantic deduplication, MCP, ChromaDB, LightRAG, PyTorch, MindSpore, BERT, Transformer, ChatGLM

Backend / Cloud: Microsoft Azure, FastAPI, REST APIs, SSE, Outlook Add-ins, Microsoft Graph, Entra ID, OAuth 2.0/OIDC, Docker, SQLAlchemy, SQLite, MySQL, Git

Data / ML: Machine Learning and Deep Learning Algorithms, Pandas, NumPy, Matplotlib, Power BI, ETL, feature engineering, model evaluation